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RESULTS

Results

01 · RANDOM EFFECTS

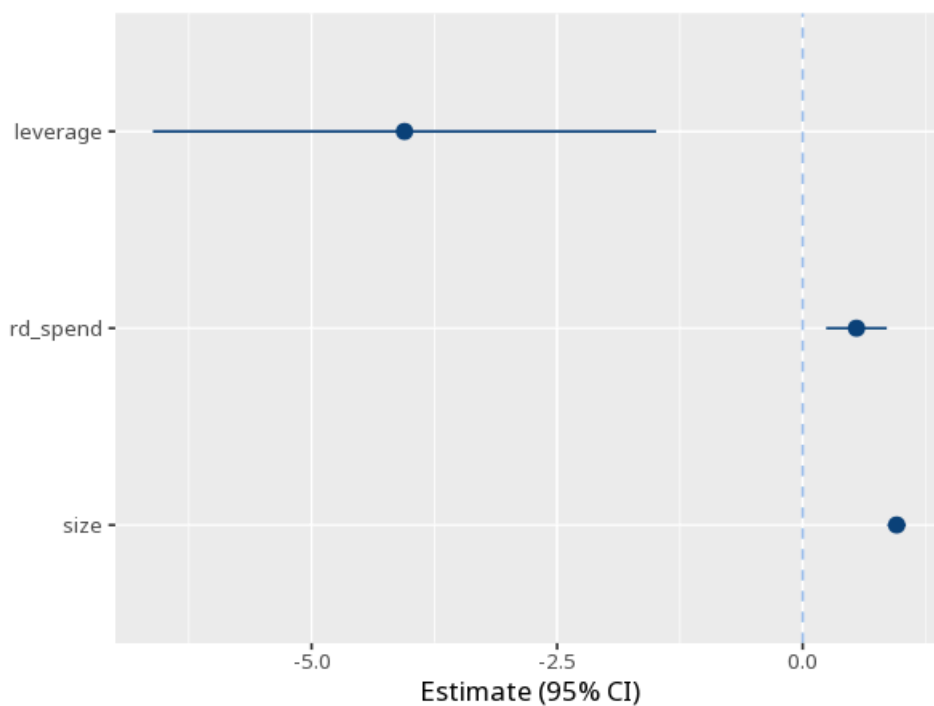
panel regression, random entity effects

Table 1. Coefficients

	(1)
(Intercept)	−0.77 (0.28) [−1.31, −0.22]
leverage	−4.05 (1.29) [−6.62, −1.49]
rd_spend	0.55 (0.16) [0.24, 0.86]
size	0.96 (0.05) [0.86, 1.05]
Num.Obs.	96
N entities	12
R ²	0.98
R ² Adj.	0.98
Within R ²	.91
Between R ²	1.00
Overall R ²	.98

standard errors are clustered by entity. unlike fixed effects, time-invariant predictors can be retained. `summary(plm)` returns a single R² / adj. R² above; within/between/overall R² (Stata xtreg convention) are also reported below. Breusch–Pagan LM test (RE vs pooled OLS): $\chi^2(1) = 0.07$, $p = .792$ — a low p favours the random effects over pooled OLS. Swamy–Arora $\theta = 0.04$ (variance components: idiosyncratic 0.25, between 0.00).

Figure. Coefficients



Understanding the numbers

B. The estimated effect of a predictor, treating entity differences as random rather than fixed - only trustworthy if a Hausman test favors random effects. Here, $B = -4.05$.

R². plm's own overall model R² for the random-effects fit. Here, $R^2 = 0.98$.

R² Adj.. That R², adjusted for the number of predictors. Here, adjusted $R^2 = 0.98$.

Within R². How well the predictors explain variation WITHIN each entity over time (Stata xtreg convention). Here, within $R^2 = .91$.

Between R². How well the predictors explain differences in the ENTITY-level averages. Here, between $R^2 = 1.00$.

Overall R². How well the predictors explain the outcome using the raw, untransformed data. Here, overall $R^2 = .98$.

N. The total number of entity-period observations used to fit the model. Here, $N = 96$.

N entities. The number of distinct entities in the panel. Here, $N \text{ entities} = 12$.

Breusch-Pagan LM. A test of random effects against pooled OLS - a low p favours random effects. Here, the Breusch-Pagan LM test gave $\chi^2(1) = 0.07$, $p = .792$.

How to read this test

Like fixed effects but treats entity differences as random, allowing time-invariant predictors. Read each B/p as usual — but only trust this model if the Hausman test favours random over fixed effects. Method: R plm package (Croissant & Millo, 2008). Your significance threshold (α) is 0.05.

APA template: In a random-effects model, predictor leverage gave $B = -4.05$, $p = .002$.

Statistical basis: Random-effects (GLS) panel estimator (Croissant, Y., Millo, G. (2008). "Panel Data Econometrics in R: The plm Package." Journal of Statistical Software, 27(2), 1-43.). Full references in the exported CITATIONS.txt.

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